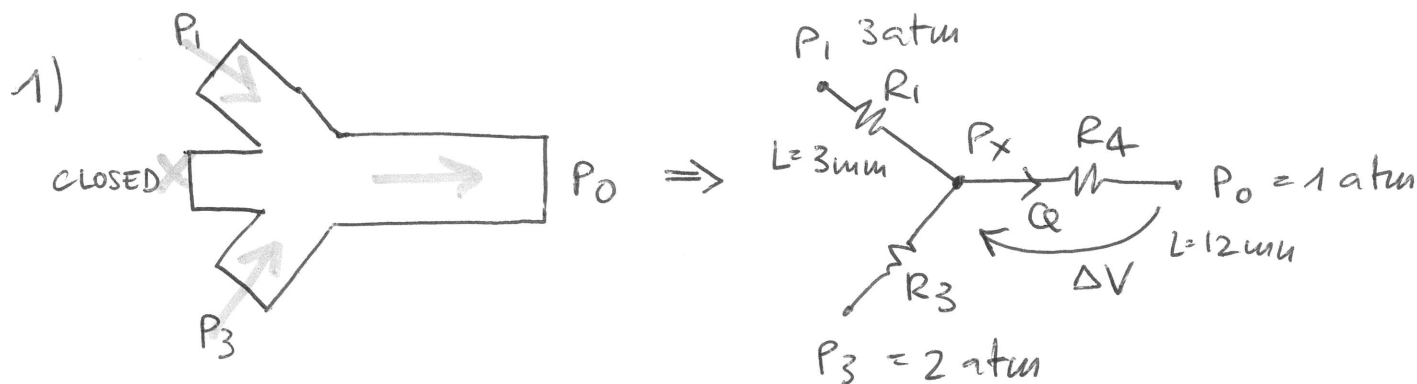




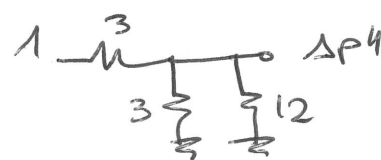
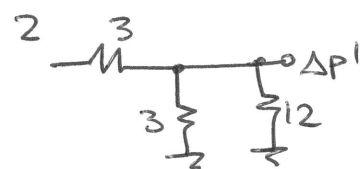
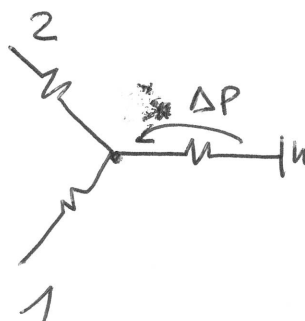
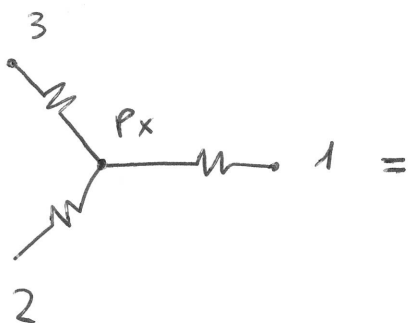
$$v_4 = \frac{Q_4}{W \cdot H}$$

$$Q_4 = \frac{\Delta V}{R_4}$$

$$\Delta V = P_x - P_0$$

$$R = \alpha \cdot L$$

find P_x with SUPERPOSITION:



$$\Delta P = \Delta P' + \Delta P''$$

$$3 // 12 = \frac{3 \cdot 12}{3 + 12} = 2.4$$

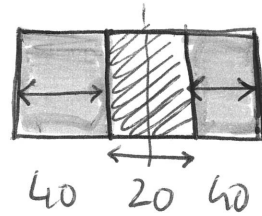
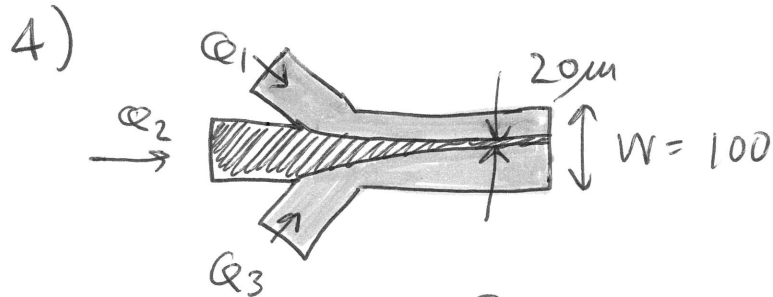
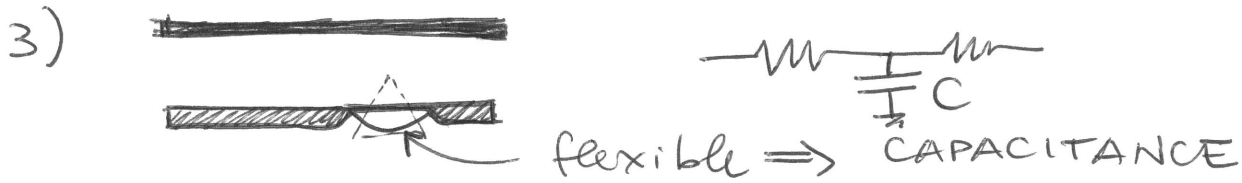
$$= 2 \cdot \frac{2.4}{3 + 2.4} + 1 \cdot \frac{2.4}{2.4 + 3} = \underline{1.33 \text{ atm}}$$

$$R_4 = \frac{12 \cdot \eta \cdot L}{(1 - 0.63 \cdot \frac{1}{2}) H^3 W} \xrightarrow{\frac{W}{H} = 2 > 1.4} = 1.68 \cdot 10^{13} \left[\frac{\text{Pa} \cdot \text{s}}{\text{m}^3} \right]$$

$$Q = \frac{1.33 \text{ atm} \cdot 10^5 \text{ Pa}}{1.68 \cdot 10^{13}} = 8 \cdot 10^{-9} \frac{\text{m}^3}{\text{s}} \quad (8 \mu\text{e/s})$$

$$v_4 = \frac{Q}{H \cdot W} = \boxed{1.6 \text{ m/s}}$$

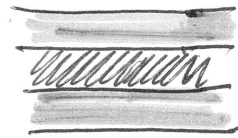
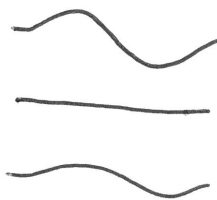
2) $\eta = \eta(T) \Rightarrow$ same model, change value of R



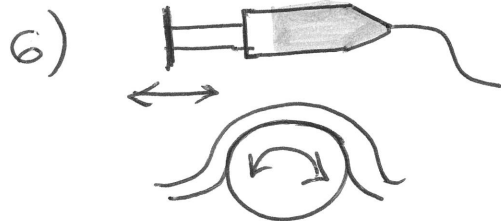
same speed: $\frac{Q_2}{2} \cdot \frac{1}{10H} = \frac{Q_1}{40H}$

$$Q_1 = Q_3 = \frac{Q_2}{2} \cdot 4 = \boxed{20 \mu e/s}$$

5) SYNCHRONIZE PUMPS



OR
LOW-PASS
FILTER

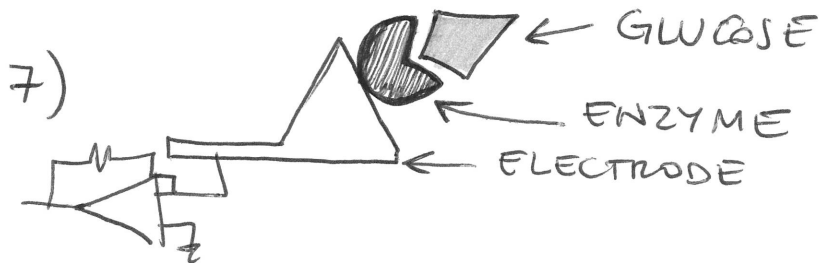


BOTH BIDIRECTIONAL

PDM IS PERMEABLE
TO GAS

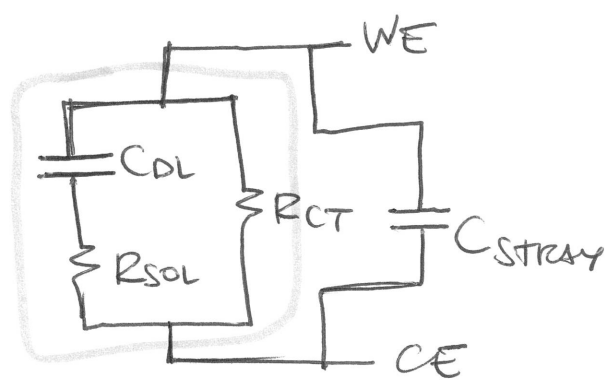
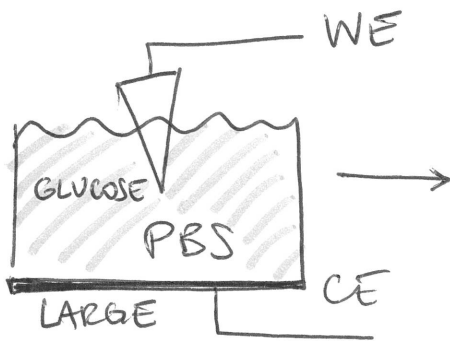


ASPIRATION IS WORSE



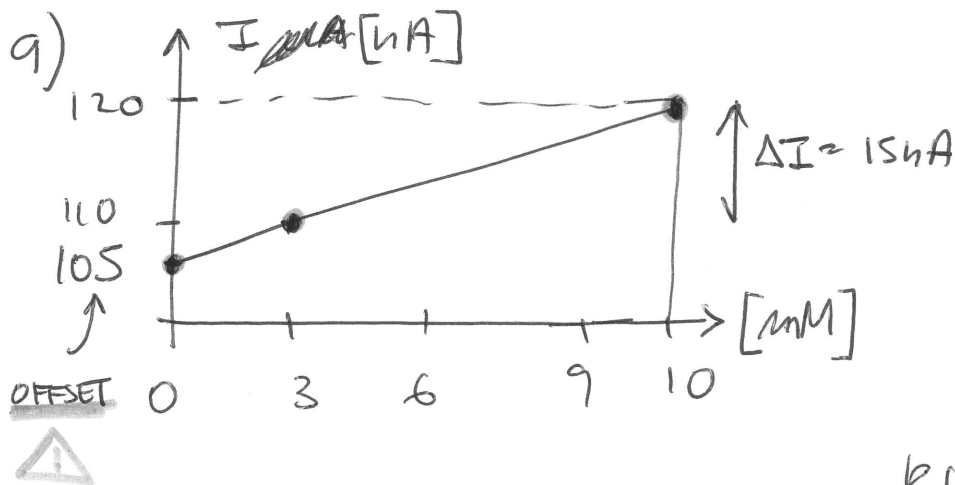
CE	WE	CE
X	X	X
CE	WE	CE

8)

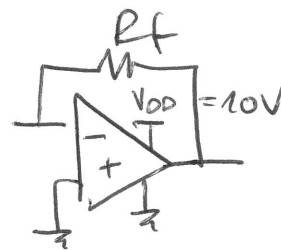


(NEGLECT C_{DL} of $CE \gg WE$)

- R_{SOL} = depends on distance from CE and geometry
- R_{CT} = depends on REDOX
- $C_{DL} = 500 \mu m^2 \cdot 0.1 \frac{pF}{\mu m^2} = 50 pF$
- C_{STRAY} = depends on parasitic coupling



$$I_{max} = 120 nA$$



$$R_f = \frac{V_{DD}}{I_{max}} = \frac{10V}{120 nA}$$

$$= 83 M\Omega$$

(COULD TAKE SOME MARGIN)

10)

$$BW = \frac{1}{2\pi R_f \cdot C_f} = 10 kHz$$

$C_{par} = 0.2 pF$

11)

$$\sigma_{RMS} = \sqrt{\left(\frac{50 fA}{\sqrt{Hz}}\right)^2 \cdot \frac{\pi}{2} \cdot 10 kHz} = 62 pA_{RMS}$$

$$LOD = \frac{6.2 pA}{15 nA} \cdot 10 mM = 4.1 \mu M$$

- 12) SILICON MASTER (LITHO) → GOLD EVAPORATION
 PDMS MOLDING → POLYMER → SPIN COATING